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Computer Vision Course Project: Improving the WPILib Vision Detection System

by

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Abstract:

The FRC - First Robotics Competition, created an abstract framework called WPILib to to allow high school students who are participating in the competition to work with less complex vision systems. However, the vision system constructed by WPILib is built to be simple and compatible with weak hardware, this introduces issues like environmental noise and high latency when more complex tasks are required. Traditionally, this was solved by creative methodologies like using polarized sunglasses to reduce sun glare, or doing additional math and storing positions to compensate and improve performance, but there are ways to mitigate these issues by improving the vision system itself. With the progression of the Computer Vision course, we have been exposed to various knowledge on how to use different methodologies to build a vision system from scratch. In this project, we will be recreating the FRC vision system using the knowledge in the Computer Vision course. The FRC vision system consists of object detection, tracking and distance estimation of game pieces lying on the competition field. We will be focusing on the detection of specific game pieces and improving it by implementing **canny edge detection** and *cross correlation in fourier domains* **least squares fitting**. The goal of this project is to improve the performance of the vision system by applying the aforementioned solutions, and achieving a more robust and efficient vision system regardless of hardware limitations, and achieve less false positives and improve performance.

1 Introduction

Using **canny edge detection** and **least squares fitting**, the goal for this project is to improve upon the issues and performance of the original FRC vision system without using hardware-reliant methods like neural networks or machine learning, while learning how to apply the techniques taught in the course.

2 Methodologies

During the recreation of the vision system, various issues were encountered while using the initial planned fourier transforms and cross correlations. As the game pieces are hard to obtain to allow testing of the vision system, similar shaped objects like a basketball were used to test the system. However, the detection failed and did not produce any meaningful results. In search of a solution, the canny edge detection algorithm was superior for this type of detection as it was able to detect the outline of the basketball well, and only required filtering of the noise. This was done by applying a mask that filtered out HSV values that were not similar to the basketball and apply the canny edge detection algorithm, and this was able to filter out most of the noise and only leave the basketball. In order to detect the basketball, least squares was used to fit an area to the basketball. Given that the equation of a circle is

$$(x-h)^2 + (y-k)^2 = r^2$$
$$x^2 - 2hx + h^2 + y^2 - 2ky + k^2 = r^2$$
$$x^2 + y^2 = 2hx + 2ky + (r^2 - h^2 - k^2)$$

This is equivalent to $Ax = b$ where A is $\begin{bmatrix} 2x_1 & 2y_1 & 1 \\ 2x_2 & 2y_2 & 1 \\ \vdots & \vdots & \vdots \\ 2x_n & 2y_n & 1 \end{bmatrix}$, x is $\begin{bmatrix} h \\ k \\ r^2 - h^2 - k^2 \end{bmatrix}$ and b is $\begin{bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_n^2 + y_n^2 \end{bmatrix}$, where (x_i, y_i) are the coordinates of the edge points detected by the

canny edge detection algorithm. By solving for x using least squares, we can find the values of h, k and $r^2 - h^2 - k^2$, and from there we can find the radius of the circle and the center of the circle, which is the position of the basketball.

$$\begin{aligned} Ax &= b \\ A^T Ax &= A^T b \\ (A^T A)^{-1} A^T Ax &= (A^T A)^{-1} A^T b \\ x &= (A^T A)^{-1} A^T b \end{aligned}$$

Using the function findContours, we are able to find the contours of the detected edges that aren't filtered by the HSV mask. This allows for the least squares to be able to use the points on the contour to fit a circle wrapping the basketball.

3 Results

The vision system was able to detect the basketball in various lighting conditions and was able to filter out most of the noise, however there were still some false positives. We used two items to test the system, a basketball and an image of the game piece. The system was able to detect both clearly as shown.

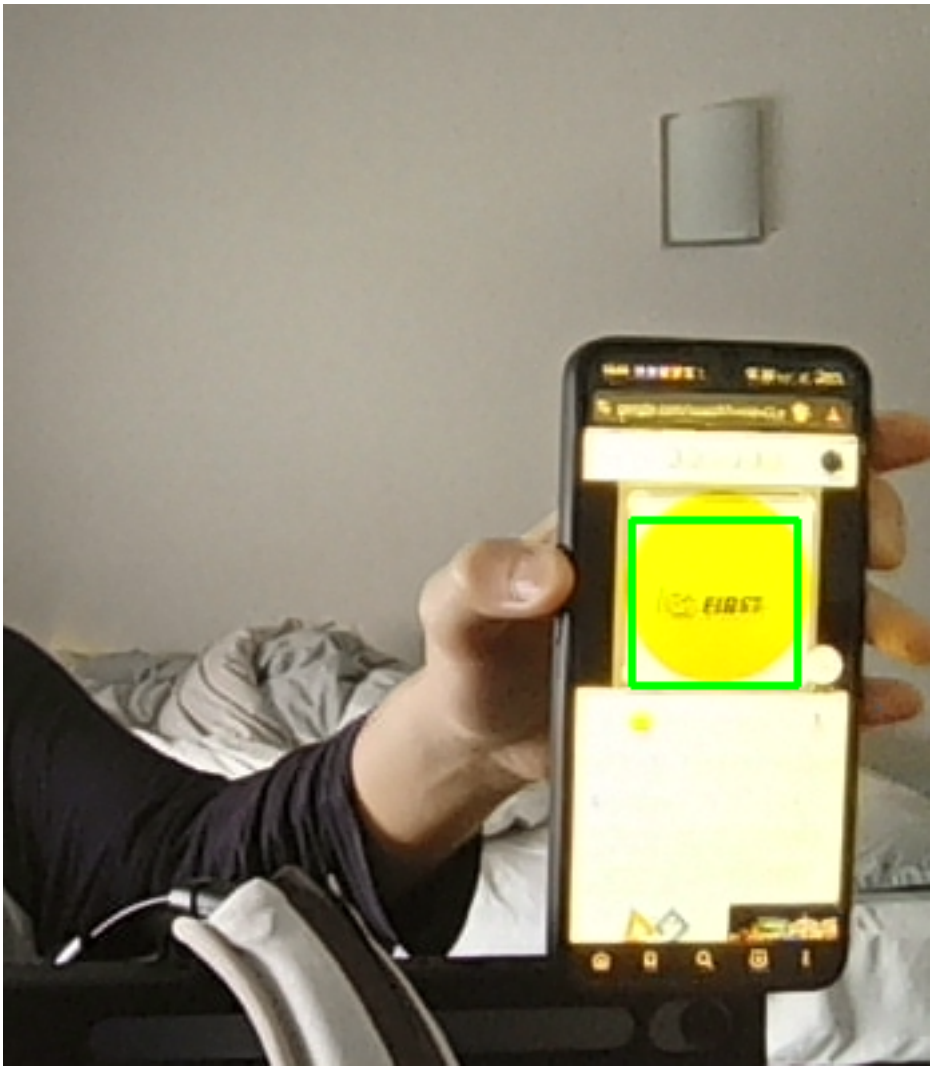

Figure 1: The image shows the detection algorithm successfully detecting an image of the game piece on a cell phone, even with glare and poor lighting

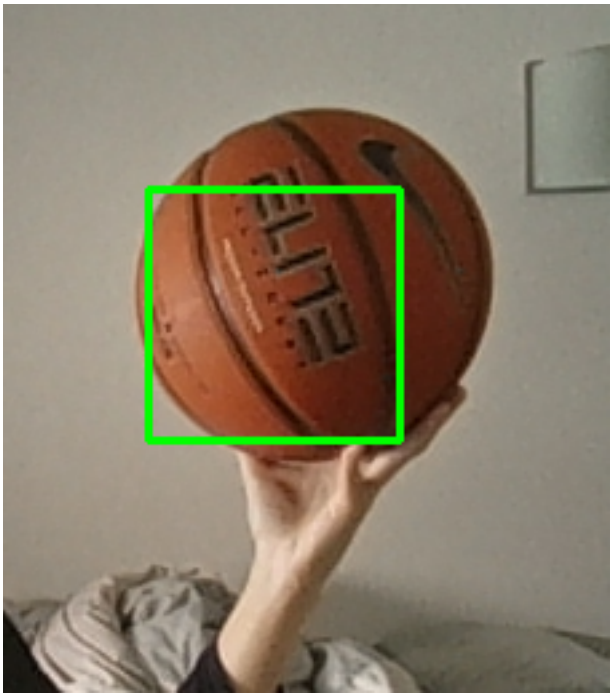

Figure 2: The image shows the detection algorithm successfully detecting a physical basketball.

4 Conclusions/Findings

The vision system was successfully recreated using the canny edge detection algorithm and least squares to fit lines to the detected edges and use those lines to find the unknowns and find the position of the basketball. From the work done, we were able to learn that the fourier transforms and cross correlations were not suitable for this, and more suited for repeated patterns and textures, while using the canny edge detection algorithm with least squares was more robust and accurate for this task. Even though the basketball and the game piece are both spheres, any differences in the shape and size of the basketball and the game piece to the template image from the perspective of the camera, depending of the angle of the texture etc, can cause fourier to fail. Fourier transforms and cross correlations measure the similarity between two signals, but is easily influenced by things like different angles and lighting conditions, while the canny edge detection algorithm with an HSV mask to isolate the Hue and ignore the brightness and saturation is

able to detect the edges of the basketball regardless of the angle and lighting conditions, and least squares is able to fit a circle to the detected edges to find the position of the basketball. Overall, the vision system was successfully recreated and improved upon by using the canny edge detection algorithm and least squares, and was able to improve upon the original vision system by applying techniques that were taught in the Computer Vision course, and was able to achieve a more robust and efficient vision system regardless of hardware limitations.

4.1 Code Repository

The code for this project can be found at the following GitHub repository:

`https://github.com/KureYv/Ball-Detection`